ACI UNIVERSAL PAYMENTS℠

RELEASE NOTES

ACI Proactive Risk Manager™ Release 9.0.2.0

September 18, 2020

ANY PAYMENT,
EVERY POSSIBILITY.®

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TABLE OF CONTENTS

ABOUT THIS PUBLICATION

This document provides a summary of the recent changes made to ACI Proactive Risk Manager (PRM).

Audience

This document is intended for managers and staff responsible for consideration and/or implementation of the PRM service pack.

Documentation and support

A HELP24 support portal account allows you to access all online product documentation, and view or log support cases through the HELP24 database. To apply for an account, contact your ACI Worldwide account manager.

https://www.aciworldwide.com/support

1. GETTING STARTED

1.1 Terminology

Open Investigation – refers to an Alerts that is not consumed immediately upon closing the Review Screen and can be picked up later for review.

1.2 Prerequisites or dependencies

Refer to the *PRM Hardware and Software Requirements Guide* and the *AR Hardware and Software Guide*.

1.3 Licensing

Refer to the “PRODUCT ENHANCEMENTS” section of this document.

1.4 Transaction based pricing

No changes in this release.

1.5 Deprecation information

No changes in this release.

1.6 Installation changes

Refer to the following documents details:

- Core Installation Guide
- AR – Web UI – Installation Guide
- Online Extract Interface (OEI) Installation Guide

1.7 Upgrade and rollback options

For a PRM upgrade, you must upgrade the database and reinstall the AG and UI.

Version	Upgrade support	Rollback support
8.2.1.0	Yes	No
8.3.x	Yes	No
8.4.x	Yes	No
8.5.x	Yes	No
8.6.x	Yes	No
8.7.x	Yes	No
8.8.x	Yes	No
9.0.x	Yes	No

1.8 Test environment

The test environment should mirror the production environment but at a smaller scale. The database version, operating system version, and Java JVM version should be the same in both the User Acceptance Test environment and Production environment.

1.9 Release artifacts and their dependencies

Artifacts included in release. Actual artifacts included in your distribution will be based on licensed features, database, and webserver. For package names, descriptions, and file names, see the *Appendix*.

2. PRODUCT ENHANCEMENTS

The enhancements included in this release are described below.

2.1 Web UI and Workflow enhancements

2.1.1 Anti-Money Laundering (AML) enhancements

This release builds on the AML enhancements that were introduced in PRM 9.0.1.0.

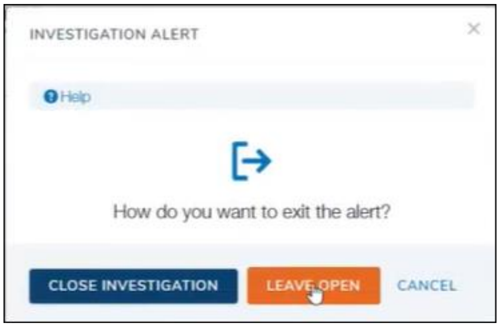
In PRM 9.0.2.0, a third value is added to the drop-down list in the Alert Property field on the Standard and Profile Rule Editor screens, the Standard and Profile Queue Editor screens, and the Category Editor screen. The new value is “Open Investigation and Archive Alerts.” Like the other values in the Alert Property field, the new value does not affect how alerts opened in the Swing user interface are handled; it affects only the alerts opened from the A&R Web user interface. Alerts opened in the Swing user interface continue to be discarded when closed, regardless of the Alert Property setting.

On the Pending Alerts page in the A&R Web user interface, when a user selects a queue with the Alert Property of “Open Investigation and Archive Alerts,” the tab that appears above the grid has a turquoise line beneath its label. When the user opens one of the alerts from that queue, the alert is moved to the ALERTS_TRACKING table with a Status of 4 (Open Investigation). In the Action History section of the Review page, the words “Opened Investigation” appear in the Action Name field.

A new column was added to the grid on the Pending Alerts page for “Discard and Archive” queues and categories. The column is labeled Time since Alerted. It indicates how much time has passed since the alert was placed in the queue. The format is number of days, followed by number of hours, followed by number of minutes (for example, 364d 4h 27m).

Three new columns were added to the grid on the Pending Alerts page for “Open Investigation” queues and categories. The new columns are labeled Status, Owner Name, and Time since Alerted. The Status column contain the value Open investigation if the alert has already been opened and was left open for investigation; otherwise, it contains the value Pending. If the Status is Open investigation, the Owner Name column contains the name of the user who owns the investigation; otherwise, it is N/A. The Time since Alerted column indicates how much time has passed since the alert was placed in the queue. The format is number of days, followed by number of hours, followed by number of minutes (for example, 364d 4h 27m).

When a user tries to close an alert that was opened from an “Open Investigation” queue or category, the following pop-up window is displayed:



If the user clicks LEAVE OPEN, the pop-up window and the Review Screen closes. When the user opens the Pending Alerts page and selects the queue or category from which the alert was opened, the user will see that the alert is still listed. The Status column contains the value Open investigation, and the Owner Name column contains the user's name.

If the user clicks CLOSE INVESTIGATION, the pop-up window, and the alert closes. When the user opens the Pending Alerts page and selects the queue or category from which the alert was opened, the user will see that the alert is no longer listed; it has been discarded.

2.1.2 Transaction Statistics widget

The name of the Transaction Volume Overview widget has been changed to Transaction Statistics.

In prior releases, the widget could display only one type of KPI (Txn Start to End). With PRM 9.0.2.0, a user can choose one or more KPIs depending on which are available in the environment for the selected Analytic Gateway(s). The EDIT WIDGET window that displays when a user selects the Transaction Statistics in the Report Type field now contains a drop-down field labeled AG KPI name. The user can select one or more KPIs from the list in that field.

The generated Transaction Statistics report has a new column labeled KPI Name.

The following list describes the seeded KPIs that are measured:

- 0-TxnStart2End - the total (start to end) CPU time for transaction processing.
- 1-DbNrtFeatureCalcProcCall - NRT feature calculation procedure call duration (when using USE)
- 1-DbRtFeatureCalcProcCall - RT feature calculation procedure call duration (when using USE)
- 1-DbNrtEntryProcCall - NRT entry stored procedure call duration
- 1-DbRtEntryProcCall - RT entry stored procedure call duration
- 1-DbRtuEntryProcCall - RTU entry stored procedure call duration
- 2-ScoringEngineCall - USE scoring request call duration (when using USE)
- 3-InputQueueWaitTime - Time spent by the incoming PIM messages in the input queue before being processed
- 3-ScoringResponseQueueWaitTime - Time spent by the incoming USE responses in the input queue before being processed (when using USE)
- 3-ScriptedCompositeField - Time spent to calculate the JavaScript Composite field (when using USE)
- ArchiveTransaction -Time spent saving (archiving) transactions on disk

- DataSanitizer - Time spent sanitizing transactions (converting certain pre-configured "bad" characters found in an incoming message into "good" ones)
- DataType - Time spent parsing an incoming message into a list of fields - values and data types (based on configured record source) to be passed in as parameters for a database stored procedure
- FieldCalculator - Time spent formulating composite fields via hashing and/or calculating from other fields. Typically used for producing a transaction tiebreaker.
- FilterCalculator – time spent on adding the configured filter fields to in-flight transaction data based on record source and licensed action levels
- IFRequest – time duration on preparing message data (after the formatting/transformation process) including determining execution flow based on state of data for IF, XF and PIM messages.
- NRTRules – NRT rule stored procedure duration
- TransactionValidation – Duration of field validation for accepted values based on a given record source

2.1.3 Opening alerts and dimension IDs in the Web U

When a user attempts to take any one of the following actions:

- open same dimension ID as an already opened dimension ID
- open same dimension ID as an already opened alert
- open same alert as an already opened dimension ID

the following message is displayed: "The same dimension ID is open in another browser tab. Do you want to move it to the current browser tab?" The user can click YES to move the dimension ID to the current tab or NO to take no action.

From any grid that contains dimension ID links, a user can use the middle mouse button, Ctrl + click, or right-click menu to open a dimension ID in another browser tab.

2.1.4 Bring all to current tab

A "Bring all to current tab" button is added to the upper left side of the OPEN DIMENSION ID LIST window, directly above the grid. A user can click the button to move all the dimension IDs that are opened in one or more other browser tabs to the current browser tab.

The following sample OPEN DIMENSION ID LIST window indicates that the Terminal dimension ID is opened in the current browser tab and that Merchant, Employee, and Card dimension IDs are opened in one or more other browser tabs. When the user clicks the "Bring all to current tab" button, the OPEN DIMENSION ID LIST window closes and the Merchant, Employee, and Card dimension IDs are moved to the browser tab where the Terminal dimension ID is opened.

2.1.5 Contextual Mark History tab

A new tab, labeled Contextual Mark History, is added to the Transaction grid on the Review screen (Swing UI). The difference between the Mark History tab and the new tab is that the Mark History tab displays only Marked transactions and the Contextual Mark History tab displays both Marked and Unmarked transactions.

By default, the Contextual Mark History tab is displayed in the Transaction grid on the Review screen for all dimensions (action levels). If a customer does not want that tab displayed or does not want one or more columns in that tab displayed, a root administrator can use the Screen Configuration Manager screen to prevent the tab or columns from being displayed for one or more dimensions.

As part of this enhancement, additional columns are included in the Mark History tab so that it and the Contextual Mark History tab have the same columns.

2.1.6 Auditing screen access in the Web UI

When a user accesses any of the following screens in the Web UI, that activity will be logged to the SystemAudit table:

- **Dashboard** - The system will audit access to the screen when the user clicks on the menu entry. It will also audit the access to the screen if it is loaded by default by the system on login.
- **Review** - The system will audit the opening of a dimension ID as alert or review.
- **Search** - The system will audit access to the screen when the user clicks on the menu entry.
- **Pending Alerts** - The system will audit the opening of a dimension ID as alert or review.
- **Point of Compromise review** - The system will audit the opening of a dimension ID as POC alert.
- **Activity History Report** - The system will audit access to the screen when the user clicks on the menu entry.
- **Forwards & Reminders** - The system will audit access to the screen when the user clicks on the menu entry.

The System Audit Report, which is generated in the Swing UI, can be configured to report the screen access in the Web UI.

2.1.7 Activity History panel

Prior to this release the Activity History was a tab in the top part of the Review page in the Web UI. Now, it and the Demographic Data are two panels that are both visible when the Review screen opens. Between them is a vertical bar that can be slid to the left or right to alter the width of each panel. By default, each panel takes up half of the display area. However, when a user moves the bar, the new configuration (split) is saved for the next time the user opens the Review page.

2.2 Desktop UI enhancements

2.2.1 Login details

After a user logs in to PRM in the Swing UI, the following information is displayed in a series of fields along the bottom of the screen:

- The organization the user entered in the Login screen
- The user name the user entered in the Login screen
- The date and time of the user's last successful login
- The date and time of the user's last failed login. If there have not been any failures, the word NONE is displayed in this field.
- The local date and time.
- The server date and time.

2.2.2 Real-Time Disposition Manager

The new Real-Time Disposition Manager screen enables users to add and delete customer real-time dispositions.

A real-time disposition value can be either a single alphabetic character or digit. The descriptive name associated with the value is what users see listed in the drop-down field in the upper right corner of the Rule Real Time Editor (Add) screen.

Custom real-time dispositions cannot be deleted if they are used in one or more rules. The rule or rules that use the custom real-time disposition must be deleted before the customer real-time disposition can be deleted.

The permission required for access is Real-Time Disposition Manager. It is assigned by default to the Root Administrator role, but it can be assigned to any custom role.

The add and delete actions are audited in the SystemAudit table. The actions can be included in the System Audit Report by selecting the Real-Time Dispostion Add and Real-Time Disposition Delete subtypes, which are listed under the Configuration node on the report configuration screen.

The Real-Time Disposition Manager screen is accessed from the Config menu (**Config > Editors**).

2.2.3 Profile Jobs Management

The Profile Jobs Management feature enables users to manage Standard Profile Rules jobs and review their progress, through the user interface. It also provides the option to run multiple processes in parallel to improve performance. The user and system actions that affect job processing are audited. That information can be monitored through the user interface, as well as included in the System Audit Report. This feature replaced the database processes used for Standard Profile Rules jobs prior to Release 9.0.2.0.

There are two screens dedicated to the Profile Jobs Management feature. They are the Standard Profile Rules Job Status and the Standard Profile Rules Job Audit.The Standard Profile Rules Job Status screen enables users to take actions, such as starting and stopping jobs, and to monitor the progress of user actions and those taken by the system. PRM audits the actions taken by users and the system on Profile Rules jobs and displays the audit information in a grid on the Standard Profile Rules Job Audit screen. The screens are accessed from the Config menu (**Config > Profile Jobs Management > Standard Profile Rules Job Status** and **Config > Profile Jobs Management > Standard Profile Rules Job Audit**).

The permissions required to access the screens have the same names as the screens: Standard Profile Rules Job Status and the Standard Profile Rules Job Audit. The permissions are assigned by default to the Root Administrator role.

The Profile Jobs Management feature is configured at the tenant level. The feature is configured through the Action Level Editor and the SystemConfig Table Editor. A new panel containing the Standard Profiles Job fields has been added to the Action Level Editor. The values set for those fields are stored in the new STANDARD_PROFILE_CONFIG column of the Action_Level table. Two new parameters, StdProfParallel and StdProfCronExpression, have been added to the SystemConfig table to configure this feature.

All configurations, job statuses and status changes, user interface actions, and automated actions are audited in the SystemAudit table and can be included in the System Audit Report by selecting the corresponding subtypes listed under the Standard Profiles node on the report configuration screen.

Important: If any jobs were previously scheduled from the database, they must be stopped and rescheduled from the UI. The application administrator user has access by default to all configuration and management screens for Profile Jobs Management, however any user can be granted permission.
This does not apply to the following dimensions: IP, Payee, Device ID.

2.2.4 Point of Compromise procedure

Users can now manually run the Point of Compromise (POC) calculation procedure by clicking the Start Poc Job button () in the tool bar on the Swing user interface. This function is especially helpful if a user wants to check multiple times a day for POC alerts.

The Start Poc Job button is visible in the user interface only if the user has the Poc Job permission. That permission is assigned by default to the Root Administrator, but it can also be assigned to custom roles. If the user has the Poc Job permission, but POC is not enabled, then the button will be visible but grayed out.

If the user has the Poc Job permission and POC is enabled, but the POCJobName parameter is not configured, then an error message stating that no job name is configured will be displayed when the user clicks the Start Poc Job button.

If the calculation procedure is running when a user is attempts to trigger the procedure from the user interface, a warning message will be displayed informing the user that the action is not available.

If the user has the Poc Job permission, POC is enabled, the POCJobName parameter is configured, and the procedure is not currently running, then the POC procedure will be run. The status will be displayed and updated until the job is completed. Once the procedure has completed, a confirmation message will be displayed.

The user's actions are audited in the SystemAudit table and can be included in the System Audit Report by selecting the Start Poc Job subtype listed under the Screen Access node on the report configuration screen.

Once the POC Alerts are generated, if the users choose to not process the Alert once is retrieved from the Alerts table, a new action will audited - Skip Alert which will also be visible in the Action History from the Review Screen of the Alerted Terminal or Retailer.

The POC calculation procedure was also enhanced to check the type of transaction that were done at the Point for Fraud and the type of Point of Compromise to Alert.

The new procedure takes in consideration new configuration parameters added in the Config table: PocCalcProcCNP, PocCalcProcCNP, PocCNPColumn and PocCNPValue .These parameters can be used to differentiate between Card Present and Card Not Present transactions.

Parameter	Description
PocCalcProcCNP	CP/CNP calculation option for Point of compromise. Valid values for LCFGVALUE: 1 – enabled, 0 – disabled Disabled: CP/CNP option is not taken into consideration at POC calculation Enabled: CP/CNP option is taken into consideration at POC calculation Default: 0
PocCNPCalc	CP/CNP calculation option for Point of compromise when PocCalcProcCNP is enabled (1) Valid values for LCFGVALUE: 1 – CNP, 0 – CP CNP option (PocCNPCalc set to 1): Only CNP transactions are taken into consideration both at the point of fraud and the point of compromise. This indicates that transactions must meet criteria entered for PocCNPColumn (eg: SD_PT_SRV_ENTRY_MDE) and its configured values at PocCNPValue (eg: 01, 02, 03...etc) Example: Both Point at fraud and Point of compromise must include transactions which have the exact configured values (PocCNPValue: 01, 02...etc) for the configured column (PocCNPColumn: SD_PT_SRV_ENTRY_MDE or SD_PT_SRV_COND_CDE) CP option (PocCNPCalc set to 0): Only CP transactions are taken into consideration at the point of compromise. This indicates that transactions must meet criteria entered for PocCNPColumn (eg: SD_PT_SRV_ENTRY_MDE) and exclude configured values at PocCNPValue (eg: 01, 02, 03...etc) Example: Point of compromise must include transactions of accounts that exclude configured values (PocCNPValue: 01, 02…etc) for the configured column (PocCNPColumn: SD_PT_SRV_ENTRY_MDE or SD_PT_SRV_COND_CDE) Default value: 0
PocCNPColumn	The column name for CP or CNP calculation for the POC functionality: SD_PT_SRV_ENTRY_MDE (The transaction entry mode. The first two bytes of this column indicate how the transaction entered the system) or SD_PT_SRV_COND_CDE (Indicates a special condition that exists at the time a transaction is initiated) Default value: SD_PT_SRV_ENTRY_MDE
PocCNPValue	Comma delimited values of configured PocCNPColumn used at POC calculation process.

2.3 Licensing enhancements

Licensing for the Travel Notification feature is implemented for the Swing UI, Web UI, and inbound services.

When the license for the Travel Notification feature is enabled, one of the ways a Travel Notification can be added to a dimension ID is by sending a request message using the Travel Notification service. When a user opens a dimension ID to which a Travel Notification has been added through the service, the user will see the Travel Notification pop-up message. If the user selects the Actions tab (Swing UI) or Action History panel (Web UI), the user will see the words UI_SERVICE in the Reviewer column.

When the license for the Travel Notification feature is enabled, the Travel Notification service can also be used to send a request to edit or delete an existing Travel Notification.

When the license for the Travel Notification feature is disabled, a user will receive an error message when attempting to add a Travel Notification using the Travel Notification service. When a user opens the Review screen (Swing UI) or Review page (Web UI) in a system in which the license has been disabled, the user will see that the Travel Notification icon is grayed out. When the user hovers over the icon, the following hover text is displayed: “Unlicensed feature (Travel Notifications).”

2.3.1 License expiration messages (Web UI)

If the license has already expired, the user will receive the following message when attempting to log in: “Your license has expired. Please contact your administrator.”

If the license will expire in 90 days or less, the following message will be displayed at the top of the page each time a multi-tenant root administrator logs in: “Your license will expire in *n* days. Please contact you ACI representative to obtain an updated license key.” The color of the band on which the message appears will changes at 30 days from expiration. It can be closed by clicking the “x” on the right side.

2.4 Serviceability and Support

2.4.1 Run Analytic Gateway (AG) as a service

The flavor of Analytic Gateway which is a REST service provider can now be installed and started as a Windows service or Unix daemon. The AG installer will create and configure additional script files which can be executed by the user to install, start, stop, or uninstall AG. The process is similar to how AG was being installed as a service or daemon when it was not a REST service provider.

2.4.2 OEI automatic installer

The new OEI automatic installer brings the familiar PRM installer experience also to the OEI component, allowing a simplified deployment for PRM's outbound communication end-points, organizing their configuration based on a few defining concepts, like: the user action, portfolio(s), message format and underlying transport protocol.

2.4.3 Enable/disable tracing in all components

ACI made a change in log4j2.xml which allow the Operations staff to dynamically change the log level in all components, and the log4j.xml changes are monitored based on the interval and apply the levels without rebooting components. The same pattern is followed in all PRM components UI, AG, OEI, and Transaction generator.

```
<Configuration status="WARN"  monitorInterval="60" >

    <Properties>

        <Property name="log.level">INFO</Property>

    </Properties>
```

2.4.4 Tomcat support for Web UI

Tomcat is now supported as application server for Web UI.

2.5 Messages and services

2.5.1 BSI Notification services – PII data hashing configuration

For an **inbound PRM UI service**, the request must be enhanced to specify if the PAN is sent in clear.

A dynamic attribute will have to be populated for the riskSubject/subject (of type RiskSubjectIdentification_Type) of the request object. Its value must be "true" or true. Any other value or the absence of this dynamic attribute will default to the old behavior – considering that the sent PAN is hashed.

This is available for the request objects of services:

RiskSubjectManagement – operations RiskSubjectBlock, RiskSubjectBlockTerminate, RiskSubjectMemoAdd, RiskSubjectAnalysisExclude

TransactionRiskAnalysisManagement - operations TransactionRiskAlertClose, TransactionMarkFraud, TransactionUnmarkFraud, TransactionRiskAction

RiskPartnerSynchronizationV2 - operations StatusUpdate

For services RiskPartnerSynchronizationV2 and RiskPartnerSynchronization – operation Subscribe, the same form of dynamic attribute must be added, but to the request, not to the subjects.

For the **Analytic Gateway** , a new property must be defined in APPLCNFG for each AG if the PAN is needed to be sent in clear for the AG notifications. This property is named "risk.gateway.external.message.notify.PIIDataInClear" and its value should be true. This property is not installed by default.

This configuration will be applied to all requests initiated by AG. There is no differentiation based on request/portfolio/anything.

Insert into APPLCNFG (CnfgPropIDTx, PrcsID, CnfgPropValTx, DscrTx, LastUpdtTm, RowAccess,

EntryTemplate, ValidationTemplate) values (risk.gateway.external.message.notify.PIIDataInClear, AGProcessID, 'true', null, null, null, null,null);

Note: AGProcessID – represents the actual AG process ID

For the **Online Extract Interface** a new column SOPTIONS is present in table EXTERNAL_CONNECTION_MAP. This way the configuration can be done on finer granularity – the combination of LPORTFOLIOID, ACTION_TYPE_ID, SEXTCCTN and state.

This configuration will be considered for the outbound Notification API (PIM/REST) based services only – RiskPartnerEventNotify, RiskPartnerEventNotifyV2, RiskSubjectManagement, TransactionRiskAnalysisManagement.

By default, the value of SOPTIONS column is null.

It is a varchar column, but it will need to have a JSON format. The value for sending the PAN in clear should look like this:

{"PIIDataInClear":true}

Or

{"PIIDataInClear":”true”}

2.5.2 Data Extensibility – new module available

A new licensed tool is now available to PRM customers that provides the ability to perform message layout extensions using PRM metadata. It also facilitates applying existing Dimensions to customized layouts.

The new elements added in a custom layout can be used in Rules, Aggregates and Enhanced profiles and can be displayed in Review screen.

A few samples are available in the product distribution and a new guide has been made available: *PRM 9.0.2.0 - Data Extensibility Developers Guide.*

2.5.3 Swagger - OpenAPI services documentation

The PIM services supported by PRM, that have previously been exposed via REST are now documented using an OpenAPI and can be accessed in UI.More details are available in the *PRM 9.0.2.0 - Services Developers Guide*.

2.6 Database operations enhancements

2.6.1 Primary Key changes for SHORTWINDOW, RT_SHORTWINDOW and DETAIL tables

For SW, RT_SW and DETAIL tables Unique keys were changed to Primary keys. For Oracle this will be available after an install/update to 902. For MSSQL is available after clean install but for upgrade additional scripts must be run.

2.6.2 Event alerting for rules

Users, such as System and Root Administrators, DBAs, or rule writers, can define thresholds for rules so that the system checks against them and logs in the database whenever a rule violates the thresholds. They can also define one or multiple polling frequencies for checking thresholds. ACI recommends a maximum of 10 to be active at the same time. The user can activate or deactivate each polling.

For each polling frequency, the user can define the following:

- Event alerting rules (not to be confused with the PRM rules for capturing fraud)
- Actions in case an event is alerted

The user can define one or multiple event alerting rules for the same polling frequency.

An event alerting rule requires the information listed below. It is stored in the new RULE_OP_EVENTS_KPI table.

- Event ID
- Rule type to which the alerting applies: RT rules or NRT rules
- Threshold - A number that will be compared to the count of identified matches with the event alerting rule conditions in the polling interval

- Optionally, an additional free SQL condition.

  Example for RT rules: SD_REAL_TIME_DISPOSITION <> 'A'

  Example for NRT rules: SD_TRAN_CDE_CAT IN (31, 32, 33, 39)

The action that can be assigned for a polling frequency is to log the information in the new RULE_OP_EVENTS_LOG table.

The cleanup for the DB log will be set the same as for the DETAIL table.

By default, the following event alerting rules will be available in the system in deactivated status:

- 30 sec polling frequency: event alerting rules for RT and NRT with threshold of 5; action: log in DB
- 60 sec polling frequency: event alerting rules for RT and NRT with threshold of 25; action: log in DB
- 300 sec polling frequency: event alerting rules for RT and NRT with threshold of 150; action: log in DB
- 600 sec polling frequency: event alerting rules for RT and NRT with threshold of 500; action: log in DB
- 1800 sec polling frequency: event alerting rules for RT and NRT with threshold of 2000; action: log in DB

3. PRODUCT FIXES

The product fixes included in this release are described below.

Title	Description
UTF-8 encoding characters are not imported properly in PRM_REF_DATA	French characters are not successfully imported and visible in UI in the Lookup Data tab
Masked data is visible in clear in tool tip	Masking configuration now applies to tool tips as well
Export data unmasked - Log in System Audit	The export is now audited in SystemAudit with the following details: "Export Unmasked Value – Field Name: *“Database column name”* ,Display Name: *“UI label”*"
JSON over REST - support for all outbound services - OEI Mark	Support for REST was added for outbound messages TransactionMarkFraud and TransactionUnmarkFraud
RiskScores are not included in XF/PIM response messages	Introduced a new configuration property: risk.gateway.response.includes.score.fields that needs to be set to “true” in order to get the risk scores and reason codes present in the XF or BSI response messages. The setting needs to be done in the APPLCNFG table.

4. KNOWN ISSUES

Described below are the known issues.

1. In the specific situation described below, an alert will be consumed, regardless of the Alert Property of the queue in which it has been placed.
   - **First:** User1 opens an investigation and leaves it open from Queue1. User1 has two queues assigned to her, Queue1 and Queue2. The transaction that triggered the investigation alert is present in both.
   - **Next:** User2 picks up ownership from Queue1. He opens the investigation in the Review page and leaves it open. User2 has three queues assigned to him, Queue1, Queue2, and Queue3. The investigation alert is present in all three queues.
   - **Issue:** Once User2 opens the investigation for review, after taking ownership, the alert from Queue3 will be consumed, regardless of that queue's Alert Property.
2. In the Swing UI, no alert can be opened from Alerts Pending after the user changes the column order. After opening, processing, and closing an alert from the Pending Alerts screen, and then changing the order of columns in the grid, users receive an error message when they attempt to open a new alert from the Pending Alerts screen. The message is "Invalid or not licensed action level id: 444."
3. Any Open investigation will be automatically closed if forwarded to another user, queue or category. The same logic applies to Reminders, therefore each time a reminder is set the open investigation will close along with the Review screen. This is planned for enhancement in a future release.
4. Encoding changed for PRM communication with external systems, therefore if the installation was done with the current EBCDIC encoding this must be replaced with CP1047 when PRM 9.0.x is adopted.

i.e PrtcOptnList = '.NONMQJMSTARGET.NOREPLY.**EBCDIC**.SSL'

this should be replaced with :

PrtcOptnList = '. NONMQJMSTARGET.NOREPLY.**CP1047**.SSL'

5. UPGRADE INSTRUCTIONS

Refer to the following installation guides:

- PRM Core Installation Guide
- AR - Web UI – Installation Guide
- Online Extract Interface (OEI) Installation Guide

6. MAPPING CHANGES

The Notification API has an additional configuration option.

Message	Field	Mapping note
RiskPartnerSynchronizationStatusUpdateRq_ REV1_Type (request message)	Status	Field is mandatory. But if the value of “Code” is blank, PRM would treat it as “Failed” unless in the tenant database, the integer value for parameter “BlankNotifyStatusIsOK” is set to 1 in table SYSTEMCONFIG.
EventRiskAnalyzeRq_Type	ClientIP, ClientLanguage, AcctType, CVDPrsntFlag	Fields are now available in RT_SHORTWINDOW and can be used in the UI for rules writing
Non-Financial Transaction XF File (NonFinTran)	BIN	This field contains the prefix or BIN for the PAN or card number used for portfolio and filter identification in the Analysis and Review system. Database Name: SD_BIN GUI Screen Name: BIN

APPENDIX: RELEASE ARTIFACTS

Package name	Package description	File name
PRM-INSTALL	PRM-INSTALL	prminstall.zip
ARMI0	ARMI0	analytic_gateway.zip
ARMI0	ARMI0	AJI.DB.LoaderExtractor.zip
ARMI0	ARMI0	onlinehelp.zip
ARMI0	ARMI0	manuals.zip
ARMI0	ARMI0	xf-xsd-files.zip
ARMI0	ARMI0	thirdparties.zip
ARMI0-TOMCAT	ARMI0-TOMCAT	tomcat.zip
ARMI0-WEBSPHERE	ARMI0-WEBSPHERE	websphere.zip
ARMI0-JBOSS	ARMI0-JBOSS	Jboss.zip
ARMI0-ORACLE	ARMI0-ORACLE	nrt.zip
ARMI0-ORACLE	ARMI0-ORACLE	additional_scripts.zip
ARMI0-ORACLE	ARMI0-ORACLE	GenerateDBScripts.zip
ARMI0-MSQ	ARMI0-MSQ	nrt.zip
ARMI0-MSQ	ARMI0-MSQ	additional_scripts.zip
ARMI0-MSQ	ARMI0-MSQ	GenerateDBScripts.zip
ARRT0-ORACLE	ARRT0-ORACLE	rt.zip
ARRT0-MSQ	ARRT0-MSQ	rt.zip
ARMI1	ARMI1	analytic_gateway.zip
ARMI1	ARMI1	AJI.DB.LoaderExtractor.zip
ARMI1	ARMI1	onlinehelp.zip
ARMI1	ARMI1	manuals.zip
ARMI1	ARMI1	xf-xsd-files.zip
ARMI1	ARMI1	thirdparties.zip
ARMI1-TOMCAT	ARMI1-TOMCAT	tomcat.zip
ARMI1-WEBSPHERE	ARMI1-WEBSPHERE	websphere.zip
ARMI1-JBOSS	ARMI1-JBOSS	Jboss.zip

Package name	Package description	File name
ARMI1-ORACLE	ARMI1-ORACLE	nrt.zip
ARMI1-ORACLE	ARMI1-ORACLE	additional_scripts.zip
ARMI1-ORACLE	ARMI1-ORACLE	GenerateDBScripts.zip
ARMI1-MSQ	ARMI1-MSQ	nrt.zip
ARMI1-MSQ	ARMI1-MSQ	additional_scripts.zip
ARMI1-MSQ	ARMI1-MSQ	GenerateDBScripts.zip
ARRT1-ORACLE	ARRT1-ORACLE	rt.zip
ARRT1-MSQ	ARRT1-MSQ	rt.zip
IFOE0	IFOE0	AJI.DB.LoaderExtractor.zip
IFOE0	IFOE0	oei-app-distribution.zip
IFOE1	IFOE1	AJI.DB.LoaderExtractor.zip
IFOE1	IFOE1	oei-app-distribution.zip
DBEXT	DATA-EXTENSIBILTY	data-extensibility-distribution.zip
SIMULATOR-TOOLS	SIMULATOR-TOOLS	ag-with-tools.zip
SIMULATOR-TOOLS	SIMULATOR-TOOLS	prm-pim-client-simulator.zip
SIMULATOR-TOOLS	SIMULATOR-TOOLS	prm-pim-provider-simulator.zip
XSD-BUILDER	XSD-BUILDER	prm-xsd-builder.zip
ARMI1-MSQ-JBOSS	ARMI1-MSQ-JBOSS	ar-installer-MSQL-Jboss.zip
ARMI1-MSQ-JBOSS	ARMI1-MSQ-JBOSS	APSF.zip
ARMI1-MSQ-JBOSS	ARMI1-MSQ-JBOSS	APSF-manuals.zip
ARMI1-MSQ-JBOSS	ARMI1-MSQ-JBOSS	AR-manuals.zip
ARMI1-ORACLE-JBOSS	ARMI1-ORACLE-JBOSS	ar-installer-ORACLE-Jboss.zip
ARMI1-ORACLE-JBOSS	ARMI1-ORACLE-JBOSS	APSF.zip
ARMI1-ORACLE-JBOSS	ARMI1-ORACLE-JBOSS	APSF-manuals.zip
ARMI1-ORACLE-JBOSS	ARMI1-ORACLE-JBOSS	AR-manuals.zip
ARMI1-MSQ-WAS	ARMI1-MSQ-WAS	ar-installer-MSQL-websphere.zip
ARMI1-MSQ-WAS	ARMI1-MSQ-WAS	APSF.zip
ARMI1-MSQ-WAS	ARMI1-MSQ-WAS	APSF-manuals.zip
ARMI1-MSQ-WAS	ARMI1-MSQ-WAS	AR-manuals.zip

Package name	Package description	File name
ARMI1-ORACLE-WAS	ARMI1-ORACLE-WAS	ar-installer-ORACLE-websphere.zip
ARMI1-ORACLE-WAS	ARMI1-ORACLE-WAS	APSF.zip
ARMI1-ORACLE-WAS	ARMI1-ORACLE-WAS	APSF-manuals.zip
ARMI1-ORACLE-WAS	ARMI1-ORACLE-WAS	AR-manuals.zip
ARMI0-MSQ-JBOSS	ARMI0-MSQ-JBOSS	ar-installer-MSQL-Jboss.zip
ARMI0-MSQ-JBOSS	ARMI0-MSQ-JBOSS	APSF.zip
ARMI0-MSQ-JBOSS	ARMI0-MSQ-JBOSS	APSF-manuals.zip
ARMI0-MSQ-JBOSS	ARMI0-MSQ-JBOSS	AR-manuals.zip
ARMI0-ORACLE-JBOSS	ARMI0-ORACLE-JBOSS	ar-installer-ORACLE-Jboss.zip
ARMI0-ORACLE-JBOSS	ARMI0-ORACLE-JBOSS	APSF.zip
ARMI0-ORACLE-JBOSS	ARMI0-ORACLE-JBOSS	APSF-manuals.zip
ARMI0-ORACLE-JBOSS	ARMI0-ORACLE-JBOSS	AR-manuals.zip
ARMI0-MSQ-WAS	ARMI0-MSQ-WAS	ar-installer-MSQL-websphere.zip
ARMI0-MSQ-WAS	ARMI0-MSQ-WAS	APSF.zip
ARMI0-MSQ-WAS	ARMI0-MSQ-WAS	APSF-manuals.zip
ARMI0-MSQ-WAS	ARMI0-MSQ-WAS	AR-manuals.zip
ARMI0-ORACLE-WAS	ARMI0-ORACLE-WAS	ar-installer-ORACLE-websphere.zip
ARMI0-ORACLE-WAS	ARMI0-ORACLE-WAS	APSF.zip
ARMI0-ORACLE-WAS	ARMI0-ORACLE-WAS	APSF-manuals.zip
ARMI0-ORACLE-WAS	ARMI0-ORACLE-WAS	AR-manuals.zip
ARMI0-WEBFOCUS	ARMI0-WEBFOCUS	WebFocus.zip
ARMI0-WEBFOCUS	ARMI0-WEBFOCUS	WebFocus-manuals.zip
ARMI1-WEBFOCUS	ARMI1-WEBFOCUS	WebFocus.zip
ARMI1-WEBFOCUS	ARMI1-WEBFOCUS	WebFocus-manuals.zip
ARMI0	ARMI0	migration.zip
ARMI0	ARMI0	migrationTool.zip
ARMI1	ARMI1	migration.zip
ARMI1	ARMI1	migrationTool.zip
ARMI0-Rules_SQL	ARMI0-Rules_SQL	RulesLibrary_MSQL.zip

Package name	Package description	File name
ARMI0-Rules_ORA	ARMI0-Rules_ORA	RulesLibrary_ORA.zip
ARMI1-Rules_SQL	ARMI1-Rules_SQL	RulesLibrary_MSQL.zip
ARMI1-Rules_ORA	ARMI1-Rules_ORA	RulesLibrary_ORA.zip
ACIMB-SQL	ACIMB-SQL	AMG_PKG_MSSQL.zip
ACIMB-ORACLE	ACIMB-ORACLE	AMG_PKG_ORACLE.zip
SEBA0-NULL-ORACLE	SEBA0-NULL-ORACLE	pse-with-null-model.zip
SEBA0-NULL-ORACLE	SEBA0-NULL-ORACLE	pse-with-null-model.zip
SEBA0-NULL-ORACLE	SEBA0-NULL-ORACLE	pse-with-null-model.zip
SEBA0-NULL-MSQ	SEBA0-NULL-MSQ	pse-with-null-model.zip
SERT0-NULL-ORACLE	SERT0-NULL-ORACLE	pse-with-null-model.zip
SERT0-NULL-ORACLE	SERT0-NULL-ORACLE	pse-with-null-model.zip
SERT0-NULL-ORACLE	SERT0-NULL-ORACLE	pse-with-null-model.zip
SERT0-NULL-MSQ	SERT0-NULL-MSQ	pse-with-null-model.zip
USE00	USE00	ARLM-6.1.5.3.tar.zip
USE00	USE00	3rd_party_libs.zip
USE00	USE00	CLI-1.0.1.0.zip
USE00	USE00	StandAloneRMS-1.0.1.0.zip
USE01	USE01	ARLM-6.4.0.0-apsf.4.0.0.0.zip
USE01	USE01	3rd_party_libs.zip
USE01	USE01	CLI-1.0.2.0.zip
USE01	USE01	StandAloneRMS-1.0.2.0.zip